



# INFORMATICS INSTITUTE OF TECHNOLOGY

**Course:** Computer Science (L4)

**Module:** SD1 Programming

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# Design

## Pseudocode

START

CLASS HistogramApp

    FUNCTION \_init\_(traffic\_data, date)

        CREATE main window with title "Traffic Volume Histogram - date"

        CREATE a canvas for drawing the histogram

        STORE traffic\_data and date

        SET window dimensions and margins

        SET bar dimensions and maximum height

        CALL draw\_histogram()

        SET protocol for window close event

        START main event loop

FUNCTION draw\_histogram()

    FIND max\_value from traffic data for scaling

    CALL \_draw\_axes(max\_value)

    CALL \_draw\_bars(max\_value)

    CALL \_draw\_legend()

FUNCTION \_draw\_axes(max\_value)

    DRAW X-axis

    DRAW Y-axis

    FOR each hour from 0 to 23 DO

        CALCULATE x position for hour

        DRAW hour label on X-axis

    SET number of Y-axis labels to 5

FOR i from 0 to 5 DO

    CALCULATE y value for Y-axis label based on max\_value

    DRAW y label

DRAW title at the top of the canvas

FUNCTION \_draw\_bars(max\_value)

    FOR each hour and corresponding traffic data for both junctions DO

        CALCULATE x position for the hour

        CALCULATE height for Junction 1 bar based on max\_value

        DRAW Junction 1 bar

        DISPLAY numerical label above Junction 1 bar

        CALCULATE height for Junction 2 bar based on max\_value

        DRAW Junction 2 bar

        DISPLAY numerical label above Junction 2 bar

FUNCTION \_draw\_legend()

    DRAW legend for Junction 1 (red)

    DRAW legend for Junction 2 (green)

FUNCTION on\_close()

    PRINT "Closing the histogram window."

    DESTROY the main window

FUNCTION main\_program()

    WHILE True DO

        PRINT "Please enter a date for traffic data (format: YYYY-MM-DD):"

        user\_date = CALL valid\_date()

```
file_name = "traffic_data" + user_date + ".csv"
```

```
result = CALL process_csv_data(file_name)
```

```
IF result is NOT empty THEN
```

```
    CALL display_outcomes(result[0])
```

```
    CALL save_results_to_file(result[0])
```

```
    data = {
```

```
        "Hanley Highway/Westway": CONVERT result[1] to list,
```

```
        "Elm Avenue/Rabbit Road": CONVERT result[2] to list
```

```
    }
```

```
    app = CREATE HistogramApp with data and user_date
```

```
WHILE True DO
```

```
    PRINT "Do you want to select another date? (Y/N):"
```

```
    continue_choice = INPUT()
```

```
    IF continue_choice is 'N' THEN
```

```
        PRINT "Exiting the program."
```

```
        RETURN
```

```
    ELSE IF continue_choice is 'Y' THEN
```

```
        BREAK
```

```
    ELSE
```

```
        PRINT 'Please enter "Y" for Yes or "N" for No...'
```

```
END FUNCTION main_program
```

```
CALL main_program()
```

```
END
```

# Inputs

```
Enter the date as dd:42
Out of range - values must be in the range 1 and 31
Enter the date as dd:one
Integer required
Enter the date as dd:16|
```

*Figure 1 Date error handling*

```
Enter the month as mm:july
Integer required
Enter the month as mm:18
Out of range - values must be in the range 1 and 12
Enter the month as mm:06
.
```

*Figure 2 Month error handling*

```
Enter the year as yyyy:one thousand
Integer required
Enter the year as yyyy:2220
Out of range - values must be in the range 2000 and 2024
Enter the year as yyyy:2024
```

*Figure 3 Year error handling*

# Outputs

## Output for date 15:06:2024

```
Enter the date as dd:15
Enter the month as mm:06
Enter the year as yyyy:2024
*****
Data file selected: traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is: 1037
The total number of trucks recorded for this date is: 109
The total number of electric vehicles for this date is: 368
The number of "two wheeled" vehicles for this date is: 401
The total number of busses leaving Elm Avenue/Rabbit Road junction heading North is: 15
The total number of vehicles through both junctions not turning left or right is: 363
The percentage of total vehicles recorded that are Trucks for this date : 11%
The average number of Bicycles per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is: 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is: 494
The total number of vehicles recorded through Hanley Highway/Westway junction is: 543
Vehicles recorded through Elm Avenue/Rabbit Road as scooters: 10%
The number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 39
The most vehicles through Hanley Highway/Westway were recorded between: 18:00 and 19:00
The number of hours of rain for this date is: 0 hours
Closing the histogram window.
Do you want to select another date? (Y/N): |
```

Figure 4 Output for dare 1/06/2024

## Traffic Volume Histogram - 15062024

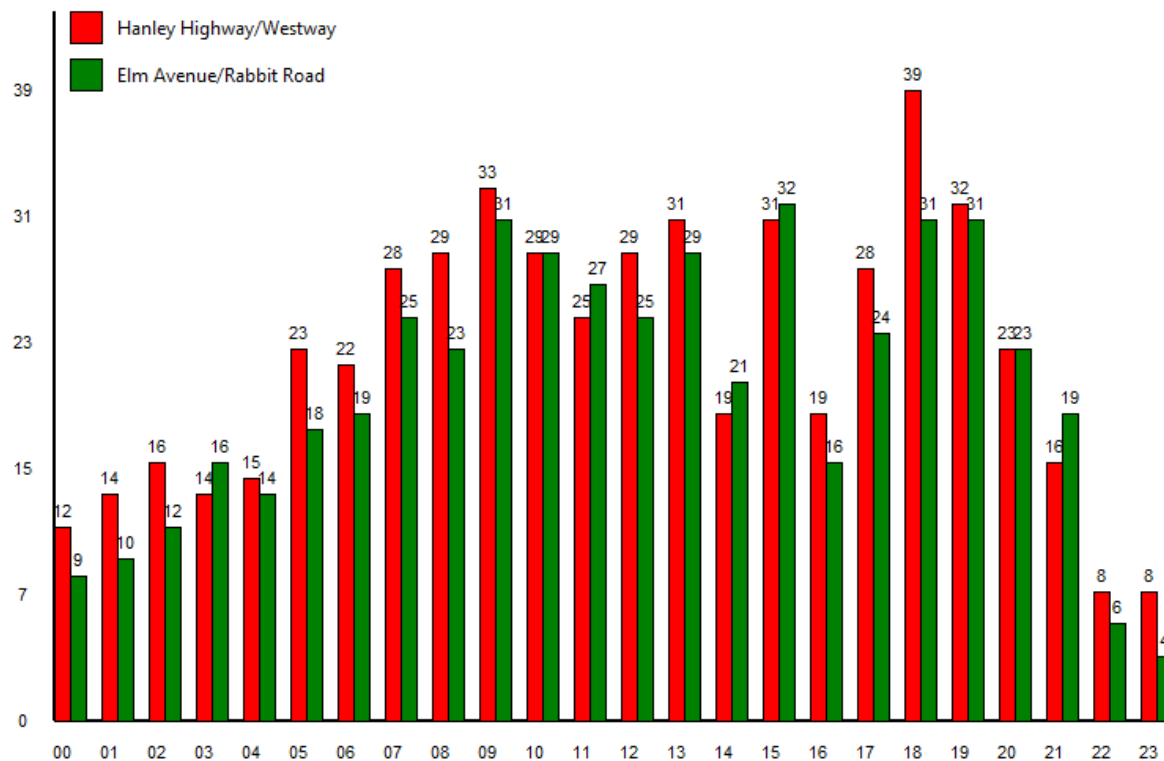


Figure 5 Histogram for 15/06/2024

```

Enter the date as dd:16
Enter the month as mm:06
Enter the year as yyyy:2024
*****
Data file selected: traffic_data16062024.csv
*****
The total number of vehicles recorded for this date is: 101
The total number of trucks recorded for this date is: 11
The total number of electric vehicles for this date is: 29
The number of "two wheeled" vehicles for this date is: 29
The total number of busses leaving Elm Avenue/Rabbit Road junction heading North is: 0
The total number of vehicles through both junctions not turning left or right is: 38
The percentage of total vehicles recorded that are Trucks for this date : 11%
The average number of Bicycles per hour for this date is : 0
The total number of vehicles recorded as over the speed limit for this date is: 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is: 52
The total number of vehicles recorded through Hanley Highway/Westway junction is: 49
Vehicles recorded through Elm Avenue/Rabbit Road as scooters: 5%
The number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 5
The most vehicles through Hanley Highway/Westway were recorded between: 01:00 and 2:00
The number of hours of rain for this date is: 3 hours
Closing the histogram window.
Do you want to select another date? (Y/N): y

```

Figure 6 Output for 16/06/2024

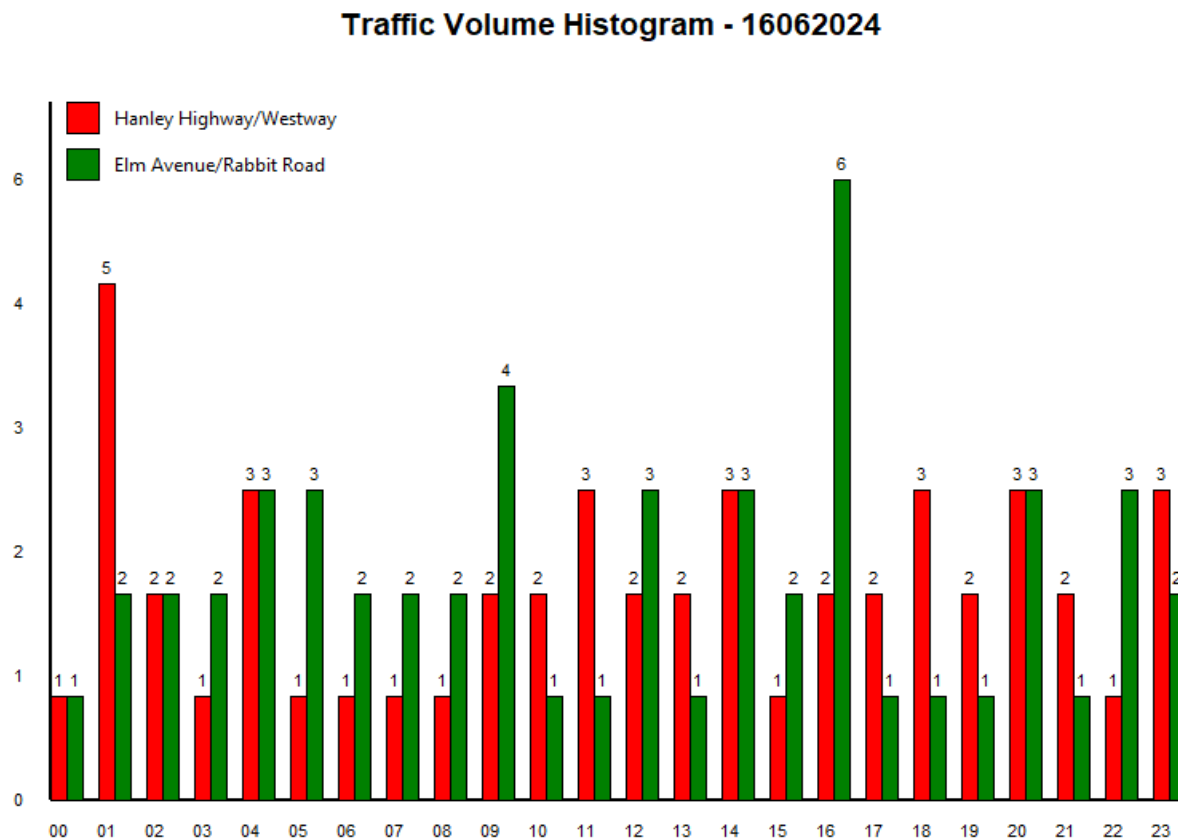


Figure 7 Histogram 16/06/2024



## Output for date 21:06:2024

```

Enter the date as dd:21
Enter the month as mm:06
Enter the year as yyyy:2024
*****
Data file selected: traffic_data21062024.csv
*****
The total number of vehicles recorded for this date is: 1334
The total number of trucks recorded for this date is: 138
The total number of electric vehicles for this date is: 442
The number of "two wheeled" vehicles for this date is: 503
The total number of busses leaving Elm Avenue/Rabbit Road junction heading North is: 19
The total number of vehicles through both junctions not turning left or right is: 494
The percentage of total vehicles recorded that are Trucks for this date : 10%
The average number of Bicycles per hour for this date is : 10
The total number of vehicles recorded as over the speed limit for this date is: 250
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is: 651
The total number of vehicles recorded through Hanley Highway/Westway junction is: 683
Vehicles recorded through Elm Avenue/Rabbit Road as scooters: 9%
The number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 71
The most vehicles through Hanley Highway/Westway were recorded between: 18:00 and 19:00
The number of hours of rain for this date is: 0 hours
Closing the histogram window.
Do you want to select another date? (Y/N): n
>>> |

```

Figure 8 Output for date 21/06/2024

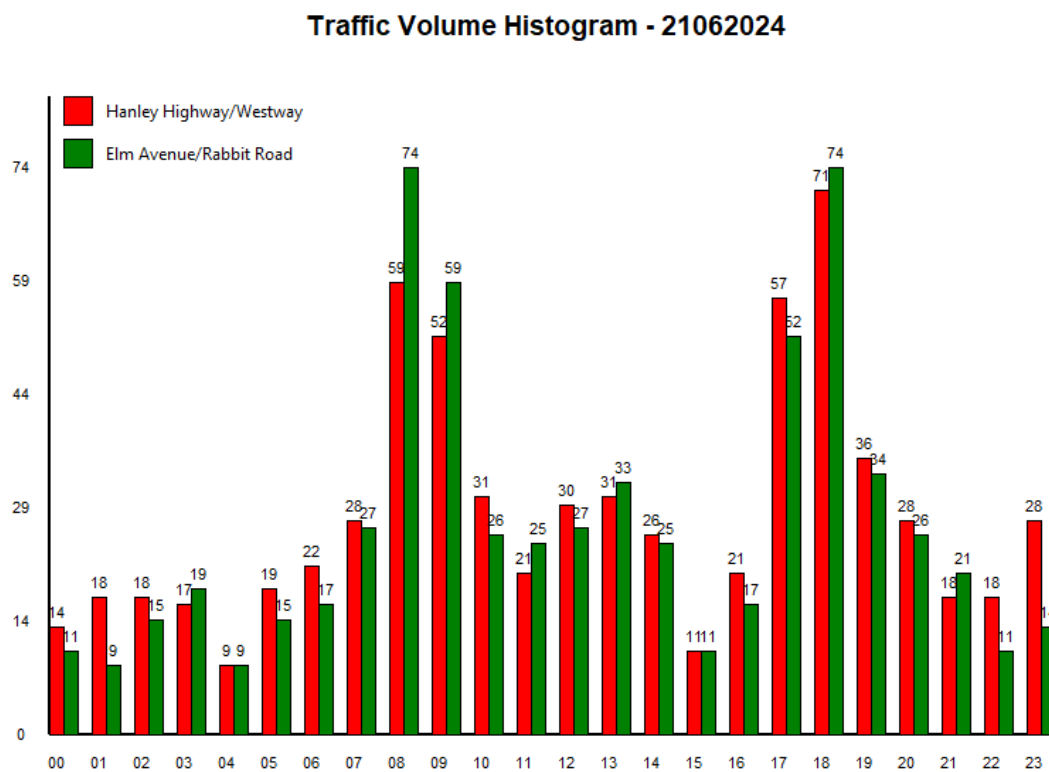


Figure 9 Histogram for date 21/06/2024

## File error handling

```
Enter the date as dd:2
Enter the month as mm:06
Enter the year as yyyy:2024
Error: File 'traffic_data02062024.csv' not found.
Do you want to select another date? (Y/N): y
```

*Figure 10 File error handling*

```
The number of hours of rain for this date is: 0 hours
Do you want to select another date? (Y/N): n
>>>
```

*Figure 11 Exit from the loop*